EVIDENCE-BASED SLEEP OPTIMIZATION

Your Sleep Blueprint

Personalized strategies backed by science - by **Dr. Martin Kawalski, MD, PhD**

1. Assess Chronotype Quiz → **2. Identify** Your Type → **3. Optimize** Targeted Protocols → **4. Track** Measure Progress

1 What's Your Chronotype? 5-MIN ASSESSMENT

Q1
If free to plan your day, when would you wake up?
- **A** Before 6:30 AM - can't sleep in
- **B** 6:30 - 7:30 AM - naturally
- **C** 8:00 - 9:30 AM - prefer late
- **D** After 9:30 AM - true night owl

Q2
When do you feel sharpest for mental work?
- **A** Early morning (6 - 9 AM)
- **B** Mid-morning (9 AM - 12 PM)
- **C** Afternoon to evening (2 - 8 PM)
- **D** Late night (after 9 PM)

Q3
How do you feel 30 minutes after waking?
- **A** Wide awake, ready to go
- **B** Alert after a few minutes
- **C** Groggy, need 30+ min to feel human
- **D** Foggy, and my sleep was light/broken

Q4
If you had to do intense exercise, when?
- **A** 6 - 8 AM
- **B** 8 - 11 AM or 3 - 5 PM
- **C** 5 - 8 PM
- **D** Whenever - hard to schedule

Q5
How would you describe your sleep quality?
- **A** Fall asleep fast, wake naturally early
- **B** Generally good, normal patterns
- **C** Hard to fall asleep before midnight
- **D** Light sleeper, wake often, rarely rested

Q6
On a free weekend, when do you naturally fall asleep?
- **A** Before 10 PM
- **B** 10 PM - 11:30 PM
- **C** Midnight - 1:30 AM
- **D** Variable - no consistent pattern

SCORE YOUR ANSWERS

- **Mostly A** — Morning Lark
- **Mostly B** — Intermediate
- **Mostly C** — Evening Owl
- **Mostly D** — Sensitive Sleeper

2 Your Chronotype Protocol PERSONALIZED TIMING

Morning Lark

Early chronotype - natural early riser, peaks before noon. ~15-25% of population. Driven by an advanced circadian phase (earlier DLMO). [11]

Sleep	**21:30 - 05:30**
Peak Focus	**06:00 - 10:00**
Caffeine Cut	**11:00 AM**

- Hardest tasks first thing - brain sharpest at dawn
- Exercise 6-7 AM to reinforce early phase
- Dim lights by 8 PM - wired to wind down early
- Don't sleep in >1h on weekends (social jet lag risk)

Intermediate Type

Neither type - solar-aligned, steady energy across the day. ~50-55% of population. The most common chronotype, well-adapted to standard schedules. [11]

Sleep	**22:30 - 06:30**
Peak Focus	**09:00 - 13:00**
Caffeine Cut	**1:00 PM**

- Morning sunlight within 30 min of waking (10+ min)
- Creative work before lunch, admin afternoon
- Exercise 8-11 AM or 3-5 PM - avoid late evening
- You adapt easily - regularity is your superpower

3 The #1 Sleep Metric You're Ignoring

2024 LANDMARK STUDY

Sleep regularity predicts mortality better than sleep duration.
A 2024 UK Biobank study (n=60,977) found the most regular sleepers had **20-48% lower all-cause mortality** - independent of how many hours they slept. Consistent bedtime and wake time outperforms hitting 8 hours on random schedules. [1]

852M	$411B	48%
ADULTS WITH INSOMNIA GLOBALLY	ANNUAL COST IN THE US ALONE	LOWER MORTALITY WITH REGULAR SLEEP

4 Universal Sleep Rules

WORKS FOR EVERYONE

YOUR IDEAL SLEEP DAY

WAKE	PEAK	WIND DOWN	SLEEP
Sunlight + Move	Deep Work	Dim + Cool	Dark + Cold

5 Bedroom Environment Hacks

OPTIMIZE YOUR CAVE

🌡️

Temperature

Cool room, warm body before bed. The drop in core temp is the trigger for sleep onset.

65-68°F / 18-20°C

🌑

Darkness

Pitch black. Even dim light (5-10 lux) during sleep reduces melatonin and disrupts glucose metabolism.

Blackout + Eye Mask

💨

Air Quality

CO_2 above 1,000 ppm reduces sleep efficiency by 1.3%. Crack a window or run a HEPA filter. [7]

< 750 ppm CO_2

🔊

Sound

Consistent background beats silence in noisy environments. White/pink noise or quality earplugs.

< 30 dB

🛏️

Bed = Sleep Only

Stimulus control: no screens, no work, no worrying in bed. If awake >20 min, get up and reset.

No Exceptions

📱

Phone Out

Charge in another room. Use analog alarm clock. Remove all standby LEDs from the bedroom.

Another Room

6 Evidence-Based Supplement Stack

BY EVIDENCE TIER

SUPPLEMENT	DOSE	TIMING	WHAT IT DOES	EVIDENCE
Magnesium L-Threonate	**144mg elemental** (~2g Magtein)	30-60 min before bed	Crosses BBB, enhances GABA signaling. 2024 RCT: +36 min total sleep time [2]	Strong
Glycine	**3g**	30-60 min before bed	Lowers core body temp via peripheral vasodilation, promotes slow-wave sleep onset [6]	Strong
L-Theanine	**100-400mg**	30-60 min before bed	Promotes alpha brain waves, reduces pre-sleep anxiety without sedation [8]	Strong
Apigenin	**50mg**	30-60 min before bed	Chamomile-derived flavonoid, mild anxiolytic via GABA-A receptor modulation [8]	Moderate
Ashwagandha KSM-66	**600mg**	With dinner	Cortisol reduction (-23%), takes 6-8 weeks for full sleep effect. 12-month safety data [5]	Moderate
Myo-Inositol	**900mg**	Before bed, 3-4x/wk	Serotonin receptor resensitization - specifically targets middle-of-night awakenings [8]	Emerging
Tart Cherry Extract	**480ml juice or capsule**	With dinner	Natural melatonin precursors + anti-inflammatory anthocyanins. Mixed results in obese [9]	Moderate
Melatonin (if needed)	**0.3 - 0.5mg only**	5-7h before target sleep	Chronobiotic phase-shifter, NOT a sedative. Physiologic dose only. Short-term use [10]	Strong

🧠 **The Huberman Stack:** Mg L-Threonate (145mg) + L-Theanine (200-400mg) + Apigenin (50mg), taken 30-60 min before bed. Add Myo-Inositol (900mg) 3-4x/week if you wake up at 2-3 AM. Start with one supplement, add another every 3-5 days to isolate effects. [8]

7 Coping Strategies

WHEN LIFE DISRUPTS SLEEP

✈️ Jet Lag Protocol

- **Eastbound:** Advance clock 1h/day for 3 days before. Morning light at destination, avoid PM light
- **Westbound:** Evening light at destination. Easier - your body naturally delays
- **Melatonin:** 0.5mg fast-release at target bedtime, max 5 days
- **Meal timing:** Eat on destination schedule immediately - gut clock shifts liver within 1-2 days
- 2025 research: full normalization takes **7+ days**, not 2 as previously thought [11]

🏭 Shift Work Survival

- **Strategic light:** Bright light first half of shift, blue-blockers last 2h before sleep
- **Anchor sleep:** Keep at least 4h at the same time daily, even on days off
- **Blackout bedroom:** Non-negotiable. Daytime melatonin levels are 50% lower without it
- **Pre-shift nap:** 20-min prophylactic nap reduces errors by 34%
- **Caffeine:** Use strategically in first half of shift only [12]

🧠 Creatine for Sleep Deprivation

- **2024 evidence:** Creatine monohydrate buffers cognitive decline during sleep loss [3]
- **Mechanism:** Increases cerebral phosphocreatine, supporting ATP recycling in the sleep-deprived brain
- **Results:** 16-29% better processing speed, ~10% better memory during 21h deprivation
- **Dose:** 3-5g daily creatine monohydrate. Best as a daily supplement, not acute rescue
- **Key:** Does NOT replace sleep - it reduces the cognitive penalty when sleep isn't possible

🧘 Middle-of-Night Waking

- **Don't check the time** - calculating remaining hours triggers anxiety loops
- **NSDR / Yoga Nidra:** 10-20 min protocol restores 60%+ of nap's cognitive benefit
- **Myo-Inositol:** 900mg before bed, 3-4x/week - specifically targets this pattern
- **20-min rule:** If still awake, get up. Read in dim light. Return when sleepy
- **Rule out:** Sleep apnea, GERD, nocturia, medication timing - see a specialist

8 Your 7-Day Quick Start START TONIGHT

9 What to Measure TRACK YOUR PROGRESS

💡 **The Zoé Sleep difference:** Generic sleep scores tell you *what happened.* We tell you **why** - and exactly what to change. Our phenotype-first approach means two people with the same "poor sleep score" get completely different protocols based on their unique biology, chronotype, and lifestyle factors.

Ready to Go Deeper?

This cheat sheet covers the fundamentals. A personalized sleep assessment identifies *your* specific bottlenecks - from circadian misalignment to hidden sleep apnea to supplement timing errors - and builds a protocol around your biology.

zoe-sleep.com

Book a personal consultation with Dr. Kawalski

martin@zoe-sleep.com • Precision Sleep Medicine • Stanford-affiliated

REFERENCES & FURTHER READING

[1] Windred DP et al. Sleep regularity index as a predictor of mortality in 60,977 UK Biobank participants. *Sleep.* 2024;47(1):zsad253.

[2] Ates M et al. Magnesium L-threonate supplementation and sleep quality: a randomized, double-blind, placebo-controlled trial. *Nutrients.* 2024;16:2669.

[3] Dworak M et al. Creatine supplementation and cognitive function during sleep deprivation. *Scientific Reports.* 2024;14:4349.

[4] Forbes SC et al. Effects of creatine on cognition: a systematic review and meta-analysis. *Frontiers in Nutrition.* 2024;11:1424972.

[5] Cheah KL et al. Effect of ashwagandha extract on sleep: systematic review and meta-analysis. *PLoS ONE.* 2021;16(9):e0257843.

[6] Bannai M, Kawai N. New therapeutic strategy for amino acid medicine: glycine improves quality of sleep. *J Pharmacol Sci.* 2012;118(2):145-8.

[7] Strøm-Tejsen P et al. Bedroom CO2 concentration and sleep quality: a field study. *Building and Environment.* 2024;247:111059.

[8] Huberman A. Toolkit for Sleep & Sleep Optimization episodes. *Huberman Lab Podcast.* 2023-2025.

[9] Barforoush NZ et al. Tart cherry and sleep: a systematic review. *Food Science & Nutrition.* 2025;13(4):e70923.

[10] Auld F et al. Evidence for the efficacy of melatonin in the treatment of primary adult sleep disorders. *Sleep Med Rev.* 2017;34:10-22.

[11] Roenneberg T et al. Munich Chronotype Questionnaire (MCTQ). *J Biol Rhythms.* 2003;18(1):80-90. | Horne & Ostberg. *Int J Chronobiol.* 1976;4(2):97-110.

[12] Sletten TL et al. Timed light exposure in shift workers: systematic review and meta-analysis. *Sleep Med Rev.* 2024;78:101990.